

Methodology – Leakage-Safe Evaluation

Correct Evaluation Flow

- 1 Load and clean raw vehicle data
- 2 Split into train and test sets
- 3 Learn budget cutoffs from **training prices only**
- 4 Apply the same cutoffs to the test set
- 5 Fit RF/CART only on `X_train`, `y_train`
- 6 Report accuracy only on `X_test`, `y_test`

Leakage Guards

- `Ex-Showroom_Price` is never used as a feature
- `Budget_Class` is removed from `X`
- `Predicted_Budget_Class` is dropped before training
- `Similarity_Score` is dropped before training
- Guard tested in `src/test_leakage.py`

Key point: Full-data fitting is used only for the final app recommendation flow, not for reporting test accuracy.